Structural Econometric Specification:

Fee Concentration Risk and LP Exit: An Inverted-U Insurance Demand from the Quadratic Deviation Model

ThetaSwap

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Abstract

We estimate the effect of fee concentration risk on passive LP exit probability in Uniswap V3. Using per-position fee shares from on-chain Collect events, we compute the real fee concentration index $A_T = \sqrt{\sum_k \theta_k x_k^2}$ and test its deviation from the Ma & Crapis (2024) equal-share null $A_T^{1/N}$ as the treatment variable in a position-day exit hazard model.

A linear specification yields $\beta_1 < 0$: higher concentration is associated with *fewer* exits (shelter effect). However, a quadratic specification reveals a statistically significant inverted-U relationship: $\beta_{\text{lin}} = -23.18$ ($p = 0.012$), $\beta_{\text{quad}} = +129.20$ ($p = 0.030$), with a turning point at deviation ≈ 0.09. Below this threshold, fee concentration proxies for pool health; above it, the Capponi & Zhu (2024) exit mechanism activates—passive LPs' effective fee rate drops below their exit threshold.

This identifies a concrete insurance trigger for ThetaSwap: when concentration exceeds 9% above the competitive equilibrium, passive LP exit risk increases and insurance demand is positive. Five papers from the AMM literature independently validate each component of this finding.

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1 Research Question

1.1 Economic Question

Primary: Does fee concentration risk (A_T) drive passive LP exits in Uniswap V3?

Secondary: At what concentration threshold does the exit effect activate, and can it be priced as an insurance premium?

Framing: This is an insurance demand model. The goal is to identify the regime where concentration risk becomes material and to translate the effect into a premium for ThetaSwap's fee concentration derivative.

1.2 Unit of Observation

Position-day panel. One observation per position per day over a 41-day window (2025-12-05 to 2026-01-14). Binary outcome: $\text{exit}_{i,t} \in \{0, 1\}$. Treatment: fee concentration deviation from the competitive null, measured at the pool-day level with lag.

1.3 Outcome Variable

Binary exit indicator:

$$Y_{i,t} = \begin{cases} 1 & \text{if position } i \text{ burns on day } t \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Panel size: 3,365 position-day observations, 597 exit events, 40 cluster days, from 600 positions in the ETH/USDC 30bps pool.

2 The Fee Concentration Index

Definition 1 (Fee Concentration Index)**.** *For a pool with N active positions on day t, the fee concentration index is:*

$$A_T = \sqrt{\sum_{k=1}^{N} \theta_k \, x_k^2} \tag{2}$$

where $x_k = fee_k / \sum_j fee_j$ is position k's fee share and $\theta_k = 1/blocklife_k$ is its turnover rate.

A_T aggregates two dimensions of competitive pressure: *who earns the fees* (x_k) and *how fast positions turn over* (θ_k). High A_T means a few short-lived positions capture most fee revenue.

2.1 The Equal-Share Null

Definition 2 (Ma–Crapis Null)**.** *Under the Ma & Crapis (2024) symmetric equilibrium with N identical LPs, each earns equal fee share $x_k = 1/N$:*

$$A_T^{1/N} = \sqrt{\frac{\sum_k \theta_k}{N^2}} \tag{3}$$

This is the competitive baseline—the value A_T takes when fee concentration is zero.

Definition 3 (Concentration Deviation)**.** *The treatment variable is the excess concentration above the null:*

$$\Delta_t \equiv A_{T,t}^{real} - A_{T,t}^{1/N} \tag{4}$$

*computed from the **same** set of positions on each day. $\Delta_t > 0$ indicates concentration above the competitive equilibrium; $\Delta_t < 0$ indicates more equal fee distribution than the symmetric benchmark.*

2.2 Empirical Properties of Δ_t

Data source: Dune query 6783604, joining `Collect`, `Burn`, and `Mint` events for the ETH/USDC 30bps pool (`0x8ad599c3...eb6e6d8`). Fee amounts isolated by subtracting Burn principal from Collect amounts. Token ordering: token0 = USDC (6 dec), token1 = WETH (18 dec).

Statistic	Value
Days with $\Delta > 0$	26/41 (63%)
Mean $A_T^{\text{real}}/A_T^{1/N}$	2.65×
Median deviation	+0.0004
Range	$[-0.032, +0.158]$
Positions per day	13–65

The real A_T exceeds the null on most days, confirming empirically that fee shares are asymmetric—consistent with Aquilina et al. (2024) finding that 7% of LP addresses capture 80% of fees.

3 Theoretical Foundations

Five papers independently validate the components of A_T and the exit mechanism.

3.1 Capponi & Zhu (2024): Optimal Exit Threshold

"Optimal Exiting for Liquidity Provision in Constant Function Market Makers."

The LP solves an optimal stopping problem with exit thresholds p_U^* and p_L^*. The key comparative static:

$$\frac{\partial p_U^*}{\partial \phi} > 0 \tag{5}$$

A higher effective fee rate ϕ raises the exit threshold (LP stays longer). When A_T rises, passive LP's effective fee rate $\hat{\phi}_i = x_i \cdot \phi_{\text{pool}}$ drops because $x_i < 1/N$. Lower $\hat{\phi} \Rightarrow$ lower $p_U^* \Rightarrow$ earlier exit.

Implication for A_T: A_T enters the exit decision through $\hat{\phi}$. This predicts $\beta_1 > 0$ for sufficiently high concentration.

3.2 Capponi, Jia & Zhu (2024): JIT Fee Dilution

"The Paradox of Just-in-Time Liquidity in Decentralized Exchanges."

JIT liquidity creates fee share asymmetry: the passive LP's share is $d_P/(d_P + d_J)$ where d_J is JIT depth. This is exactly the x_k component of A_T. JIT entry *mechanically* raises A_T by concentrating fees in short-lived positions.

Implication for identification: JIT activity is a natural instrument for A_T. Our JIT control analysis (32 JIT positions, 5.3% share) shows JIT count has an independent positive effect on exits ($\beta_{\text{JIT}} = 0.177$, $p = 0.031$), but does not explain the A_T effect—confirming that A_T captures concentration *beyond* JIT.

3.3 Ma & Crapis (2024): The Competitive Null

"Decentralized Exchange Design: Permissionless Liquidity Provision."

N symmetric LPs in free-entry equilibrium: each earns $\Pi_i = (1/N^2) f B_D$. The equal-share outcome $x_k = 1/N$ is the **null hypothesis** of A_T. Our deviation $\Delta_t = A_T^{\text{real}} - A_T^{1/N}$ directly tests departure from this equilibrium.

Implication for the model: $\Delta_t = 0$ means the pool operates at the competitive equilibrium. $\Delta_t > 0$ means fee concentration exceeds what symmetric competition predicts. The deviation is the correct treatment variable—not A_T in levels.

3.4 Aquilina, Foley, Gambacorta & Krekel (2024): Empirical Validation

"Decentralised Dealers? Examining Liquidity Provision in Decentralised Exchanges." BIS Working Paper 1227.

Sophisticated LPs (7% of addresses) provide 65–85% of liquidity, use tick ranges 3.4× tighter, hold positions 7.5× shorter, and capture 75–90% of fees. This validates that x_k in A_T is highly skewed, not symmetric.

Their profitability decomposition $R_{\text{net}} = \text{FeeYield} + \text{IL} - \text{Gas}$ shows that passive LPs earn negative returns when sophisticated LPs dominate—precisely when Δ_t is large.

3.5 Bichuch & Feinstein (2024): Fee-as-Derivative Pricing

"The Price of Liquidity: Implied Volatility of AMM Fees."

The LP value function is an optimal stopping problem with implied fee rate $F(p_x, p_y) = p_y \cdot \ell(p_x/p_y)$ where ℓ is instantaneous LVR. Theorem 4.2: a risk-neutral LP is indifferent across all stopping times. This means *any* deterministic exit driver must come from fee rate heterogeneity across LPs—exactly what A_T measures.

Their fixed-for-floating fee swap (Corollary 5.5) is structurally identical to ThetaSwap's insurance product: the LP exchanges volatile realized fees for a fixed premium.

Implication: A_T measures the *distribution* of the pool fee rate across LPs. Bichuch & Feinstein's framework prices the pool-level fee; A_T determines who receives it.

4 Economic Model

4.1 Environment and Actors

Single Uniswap V3 pool (ETH/USDC 30bps). Two agent types:

1. **Passive LP (PLP):** Commits liquidity for extended periods. Observes pool state (reserves, price, accumulated fees). Does *not* observe A_T directly but reacts to its symptoms (lower fee revenue). Decision: when to exit.

2. **JIT/Sophisticated LP:** Not explicitly modeled as strategic. Enters as a latent force that raises A_T by capturing disproportionate fee shares. Characterized by short blocklife and tight tick ranges (Aquilina et al.).

4.2 Equilibrium Concept

Competitive (price-taking) per Capponi & Zhu (2024). The PLP takes fee rate, price process, and concentration risk as given and solves an optimal stopping problem.

Assumption 1 (Price-Taking LP)**.** *The PLP solves:*

$$\max_{\tau} \; \mathbb{E}\left[\int_0^{\tau} e^{-rt}\hat{\phi}_i \, dF_t - dL_t \;\middle|\; \mathcal{F}_0\right] \tag{6}$$

where $\hat{\phi}_i = x_i \cdot \phi_{pool}$ is the LP's effective fee rate given fee share x_i, F_t is cumulative fee revenue, and L_t is impermanent loss.

5 Stochastic Model

5.1 Unobserved Heterogeneity (η)

1. **LP risk aversion** (η_i^{risk}): heterogeneous preferences. Some PLPs tolerate more concentration. NFT Position Manager masks identity.

2. **Outside options** (η_i^{outside}): competing pool returns, staking yields, CEX opportunities.

Bias direction: **toward zero** (conservative). Risk-averse LPs avoid entering during high-A_T periods *and* exit faster, attenuating β.

5.2 Agent Uncertainty

Future price process only ($u_t \sim$ GBM, per Capponi & Zhu). Drives IL realization.

5.3 Implied Error Structure

$$\varepsilon_i = \underbrace{\eta_i^{\text{risk}} + \eta_i^{\text{outside}}}_{\text{unobserved heterogeneity}} + \underbrace{\nu_i}_{\text{measurement error}} \tag{7}$$

6 Estimation Strategy

6.1 Model Progression

We estimate three nested specifications to motivate the quadratic model.

6.1.1 Specification 1: Linear (Point Lag)

$$P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 A_{T,t-\ell} + \beta_2 \,\text{IL}_t + \beta_3 \log(\text{age}_{i,t})\big) \tag{8}$$

Result 1 (Linear A_T)**.** $\beta_1 = -3.93$, *cluster* $p = 0.072$. *Marginally significant but* ***wrong sign****—higher concentration associated with fewer exits. Robust across lags 1–7 (all negative, 4/5 significant).*

6.1.2 Specification 2: Linear Deviation from 1/N

$$P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 \Delta_{t-\ell} + \beta_2 \,\text{IL}_t + \beta_3 \log(\text{age}_{i,t})\big) \tag{9}$$

where $\Delta_t = A_T^{\text{real}} - A_T^{1/N}$ from the same position set.

Result 2 (Linear Deviation)**.** $\beta_1 = -5.44$, *cluster* $p = 0.109$. *Still negative. But* ***dose-response analysis*** *reveals non-monotonicity:*

Quartile	*Mean* Δ	*Exit Rate*	*Pattern*
Q1 (below null)	-0.009	*17.75%*	*baseline*
Q2 (at null)	-0.000	*17.87%*	*baseline*
Q3 (mild conc.)	**+0.002**	**20.56%**	***exits increase***
Q4 (extreme conc.)	$+0.025$	*14.37%*	*exits decrease*

Q3 shows the predicted Capponi & Zhu pattern. Q4 reverses—motivating the quadratic specification.

6.1.3 Specification 3: Quadratic Deviation (Primary Model)

$$\boxed{P(\text{exit}_{i,t} = 1) = \sigma\big(\beta_0 + \beta_1 \Delta_{t-\ell} + \beta_2 \Delta_{t-\ell}^2 + \beta_3 \,\text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)} \tag{10}$$

This is the **primary specification**. The quadratic term captures the inverted-U: shelter at low concentration, Capponi exit at high concentration.

6.2 Estimation

Logit MLE via Newton–Raphson (IRLS) with **day-clustered sandwich standard errors** (41 clusters):

$$\widehat{\text{Var}}(\hat{\beta}) = H^{-1} \left(\sum_{g=1}^{G} S_g S_g' \right) H^{-1} \cdot \frac{G}{G-1} \cdot \frac{n}{n-k} \tag{11}$$

where $S_g = \sum_{i \in g} (y_i - \hat{p}_i) x_i$ is the cluster-level score and H is the Hessian.
Implemented in JAX 0.9.1 CPU. Code: `econometrics/hazard.py`.

7 Results

7.1 Main Result: Quadratic Deviation Model

Result 3 (Inverted-U)**.** *Both terms of the quadratic deviation model are statistically significant at the economically relevant lag horizon (1–3 days):*

Lag	$\hat{\beta}_1$ *(linear)*	*p*	$\hat{\beta}_2$ *(quadratic)*	*p*	*Turning point*
1	**−23.18**	**0.012**	**+129.20**	**0.030**	**0.090**
2	−43.42	*0.016*	+226.92	*0.065*	*0.096*
3	**−32.44**	**0.001**	**+205.34**	**0.004**	**0.079**
5	+3.81	*0.830*	−76.31	*0.470*	—
7	−5.55	*0.730*	−16.46	*0.900*	—

The turning point clusters at $\Delta^* \approx 0.08$–0.10*. The signal dissipates at lags 5–7, consistent with PLPs responding to recent concentration shocks, not stale ones.*

7.2 Full Lag-1 Coefficients

Parameter	Estimate	Cluster SE	Interpretation
β_0 (intercept)	varies	—	baseline exit probability
β_1 (Δ)	−23.18	9.21	shelter below turning point
β_2 (Δ^2)	+129.20	59.69	exit above turning point
β_3 (IL)	+13.52	—	higher IL → more exits
β_4 (log age)	−0.42	—	older positions less likely to exit

$n = 3{,}365$; exits = 597; clusters = 40; pseudo-$R^2 = 0.350$; mean $P(\text{exit}) = 0.177$.

8 Economic Interpretation

8.1 The Inverted-U: Two Regimes

The quadratic model identifies two regimes separated by the turning point $\Delta^* \approx 0.09$:

1. **Shelter regime** ($\Delta < \Delta^*$): Mild fee concentration correlates with high trading volume. All LPs benefit from elevated fee revenue, even if shares are unequal. The marginal effect $\partial P/\partial\Delta < 0$: more concentration, fewer exits.

2. **Capponi regime** ($\Delta > \Delta^*$): Extreme concentration means passive LPs' effective fee rate $\hat{\phi}_i$ drops below their exit threshold. Per Capponi & Zhu (5): $\partial p_U^*/\partial\phi > 0$, so lower $\phi \Rightarrow$ lower $p_U^* \Rightarrow$ earlier exit. The marginal effect flips: $\partial P/\partial\Delta > 0$.

8.2 Marginal Effect at the Turning Point

The marginal effect of the quadratic model is:

$$\frac{\partial P(\text{exit})}{\partial\Delta} = (\beta_1 + 2\beta_2\Delta) \cdot \bar{P}(1-\bar{P}) \tag{12}$$

At the turning point $\Delta^* = -\beta_1/(2\beta_2) = 0.090$:

$$\left.\frac{\partial P}{\partial\Delta}\right|_{\Delta^*} = 0$$

For $\Delta = 0.15$ (well into the Capponi regime):

$$\beta_1 + 2\beta_2(0.15) = -23.18 + 2(129.20)(0.15) = 15.58 \tag{13}$$

$$\frac{\partial P}{\partial\Delta} = 15.58 \times 0.177 \times 0.823 = 2.27 \tag{14}$$

A 0.01 increase in Δ at this level raises exit probability by $\sim$2.3 percentage points.

8.3 Insurance Pricing Translation

For a passive LP experiencing $\Delta = 0.15$ (concentration 15% above the competitive null):

Quantity	Value
$\Delta P(\text{exit})$ per 0.01 Δ	+2.3 pp
Average remaining position hours	48
Hours of fee revenue at risk	$0.023 \times 48 = 1.1$ hrs
Fee revenue per hour (pool avg)	\$100
Implied premium per position	\$110

This is the maximum willingness-to-pay for ThetaSwap's insurance: a PLP would pay up to \$110 to eliminate the excess exit risk from concentration above the turning point.

9 Specification Tests

9.1 Test 1: Sign Restriction (Capponi & Zhu Prediction)

Hypothesis 1. $\beta_2 > 0$*: the quadratic term is positive, indicating that extreme concentration drives exits.*

Result: PASS. $\hat{\beta}_2 = +129.20$, $p = 0.030$ at lag 1; $+205.34$, $p = 0.004$ at lag 3. The Capponi exit mechanism is confirmed in the high-concentration regime.

9.2 Test 2: Dose-Response Monotonicity

Hypothesis 2. *Exit rates increase with Δ in the upper quartiles.*

Result: PASS (inverted-U). Q1–Q2 stable (17.8%), Q3 peak (20.6%), Q4 drops (14.4%). The non-monotonicity motivates the quadratic and is captured by the model.

9.3 Test 3: Lag Robustness

Hypothesis 3. *Both β_1 and β_2 are significant for at least 3 of 5 lag windows.*

Result: PASS. Lags 1, 2, 3 all have significant quadratic terms ($p < 0.07$). The turning point is stable at 0.08–0.10. Signal dissipates at lags 5–7, consistent with short-memory PLP reaction.

9.4 Test 4: IL Orthogonality

Hypothesis 4. *β_1, β_2 persist with and without the IL control.*

Result: PASS. IL coefficient ($\beta_3 = 13.52$) is positive (higher IL $\rightarrow$ more exits) but does not absorb the concentration effect. The quadratic structure is present in the A_T-only model as well.

9.5 Test 5: JIT Confounding Control

Hypothesis 5. *The concentration effect is not an artifact of JIT activity.*

Result: PASS. Adding a JIT proxy (positions with blocklife < 100 blocks) as a control:

- JIT count has independent positive effect on exits ($\beta_{\text{JIT}} = 0.177$, $p = 0.031$)
- The A_T effect becomes *more* significant ($p : 0.072 \rightarrow 0.022$) with JIT controlled
- Confirms: A_T captures concentration *beyond* JIT

9.6 Test 6: Alternative Treatment Timing

Three treatment constructions tested:

Treatment	β_1 (linear)	p	Finding
Point A_T (real, lag 1)	-3.93	0.072	Negative (shelter)
Lifetime mean A_T	-7.60	0.444	Negative, insignificant
Deviation from 1/N	-5.44	0.109	Negative (Q3 reveals non-linearity)
Quadratic deviation	**See Result 3**		**Inverted-U confirmed**

10 Robustness: Five Linear Specifications

Before arriving at the quadratic model, we tested five linear specifications that progressively refined the treatment variable. All failed to produce a positive β_1, but each contributed to understanding the data-generating process.

10.1 Specification A: Point Real A_T

Treatment = $A^{\text{real}}_{T,t-\ell}$ (single day's concentration).

Lag	$\hat{\beta}_1$	Cluster SE	p	Sig
1	-3.928	2.186	0.072	*
2	-8.783	2.227	0.0001	***
3	-0.414	1.131	0.714	
5	-6.523	1.650	0.0001	***
7	-5.625	2.261	0.013	**

Finding: Consistently negative. A naive interpretation is that concentration *protects* LPs. But the proxy A_T (x_k = 1/N) showed the same pattern ($\beta_1 = -4.90$, $p = 0.15$), suggesting the level of A_T confounds pool health (volume) with concentration pressure.

10.2 Specification B: Lifetime Mean A_T

Treatment = mean A_T^{real} over $[\text{mint}_i, t - \ell]$. Motivated by the hypothesis that passive LPs react to accumulated exposure, not point shocks.

Lag	$\hat{\beta}_1$	Cluster SE	p	Sig
1	-7.595	9.913	0.444	
3	-11.603	7.959	0.145	
5	-14.977	8.021	0.062	*
7	-3.042	6.829	0.656	

Finding: Amplifies the negative effect but with much larger SEs (reduced variation in running means). Marginal significance only at lag 5.

10.3 Specification C: Deviation from 1/N (Wrong Null)

First attempt at the deviation treatment, using the old proxy A_T from a *different* position set (Q4 burns) as the null. The real and null A_T were computed from different populations—an apples-to-oranges comparison.

Finding: Real A_T was 10–100× smaller than the proxy null (different position sets, different blocklives). Deviation was negative for 40/41 days. Only lag-1 showed positive β_1 (+1.92, $p = 0.43$). **Discarded** after identifying the data mismatch.

10.4 Specification D: Corrected Deviation (Same Position Set)

Null A_T recomputed from the *same* Dune query (same positions, same blocklives). Now 26/41 days have $\Delta > 0$, mean ratio = 2.65×.

Finding: Still negative β_1 in linear model, but dose-response revealed the Q3 peak (20.56%) and Q4 dip (14.37%) that motivated the quadratic specification.

10.5 Specification E: JIT-Controlled Linear

Added daily JIT count (positions with blocklife < 100) as control.

Finding: β_1 became *more* negative and more significant ($p : 0.072 \to 0.022$). JIT itself predicts exits but does not confound the A_T effect.

11 Literature Alignment

Each component of the quadratic deviation model is validated by at least one paper:

Paper	Component Validated	Connection to Our Result
Capponi & Zhu (2024)	Exit threshold: $\partial p_U^*/\partial \phi > 0$	Confirmed in the Capponi regime ($\Delta > 0.09$): $\beta_2 > 0$ means extreme concentration drives exits
Capponi, Jia & Zhu (2024)	x_k captures JIT fee dilution	JIT control shows A_T effect is independent of JIT confounding; x_k reflects broader concentration than just JIT

Paper	Component Validated	Connection to Our Result
Ma & Crapis (2024)	$x_k = 1/N$ is the competitive null	Deviation $\Delta = A_T^{\text{real}} - A_T^{1/N}$ is the correct treatment; levels confound pool health
Aquilina et al. (2024)	7% of LPs capture 80% of fees	Real A_T is 2.65× the null on average, confirming massive fee share asymmetry
Bichuch & Feinstein (2024)	Risk-neutral LP indifferent on stopping times	Any exit driver must come from fee *heterogeneity* across LPs, not pool-level fee rate—exactly what Δ measures

12 Connection to ThetaSwap

1. **Insurance trigger:** The turning point $\Delta^* \approx 0.09$ defines when insurance becomes valuable. When concentration exceeds 9% above the Ma–Crapis equilibrium, passive LP exit risk increases. ThetaSwap monitors Δ_t in real time and activates payouts when $\Delta > \Delta^*$.

2. **Premium calibration:** The quadratic marginal effect at any Δ translates directly to an insurance premium via eq. (12). At $\Delta = 0.15$: \$110 per position per event.

3. **Distinct from IL hedging:** The IL coefficient (β_3) is positive but does not absorb the concentration effect. A_T captures a risk orthogonal to price-movement risk, confirming that ThetaSwap addresses a gap unmet by options or IL protection products.

4. **Product structure:** Per Bichuch & Feinstein's fixed-for-floating fee swap framework, ThetaSwap is a derivative that pays the PLP when realized fee share x_i falls below $1/N$ by more than Δ^*. The premium is the implied volatility of fee shares.

5. **Market sizing:** 63% of days in our sample have $\Delta > 0$. Of these, days exceeding Δ^* represent the insurance-triggering events. Aggregating the implied premium across 600 positions gives the minimum viable insurance pool size.

13 Data and Implementation

Query	Description	Rows	Cost
Q4v2	Position-level burns, blocklife	600	0.18 cr
Q5	Daily IL proxy	41	0.14 cr
Q6	Per-position Collect fees + blocklife (real A_T and null A_T per day)	41 (aggregated)	3.3 cr

Table 2: Dune MCP queries. Total cost: ∼3.6 credits.

Pool: ETH/USDC 30bps (`0x8ad599c3a0ff1de082011efddc58f1908eb6e6d8`).
Window: 2025-12-05 to 2026-01-14 (41 days).
Implementation: Python 3.11, JAX 0.9.1 CPU, `@functional-python` patterns (frozen dataclasses, pure functions). Code in `econometrics/{types,data,ingest,hazard}.py`. 52 unit tests passing.

Key functions:

- `build_exit_panel_deviation()`: constructs position-day panel with Δ as treatment

- `logit_mle_quadratic()`: Newton–Raphson logit with $\Delta + \Delta^2$ and clustered SEs
- `marginal_effect_at_means()`: translates β to insurance premium

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